

RAKESH VAIDEESWARAN MAHESH

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EXPERIENCE

Amazon - Artificial General Intelligence (AGI Foundations)

Sunnyvale, CA

Applied Scientist II

Dec 2023 - Present

- Building Amazon Nova multimodal foundation models across pretraining, post-training (SFT, DPO, GRPO), reinforcement learning, and model evaluation.
- Developing reinforcement learning (RL) environments and gyms to improve factuality and grounding in agentic AI systems, including task design, dataset creation, reward formulation, and scalable evaluation.
- Contributed to the research, development, and release of **Nova 1 Lite, Pro** [Dec 2024], **Nova Premier** [Apr 2025], **Nova 2 Lite, Pro** [Dec 2025] foundation models, with a focus on text and multimodal understanding capabilities across knowledge and reasoning tasks.
- Designed and implemented scalable data pipelines for curating high-quality training and validation datasets for LLM/VLM pretraining and post-training, driving improvements across model evaluation metrics.
- Identified model failure modes through systematic evaluation and error analysis, and developed targeted data and post-training interventions to improve reasoning, factuality, grounding, and multimodal understanding.
- Defined data requirements and quality criteria for third-party dataset acquisition, evaluating coverage, quality, and suitability for foundation-model pretraining and post-training.
- Collaborated across research, engineering, data, and product teams to translate experimental model improvements into production releases across the Amazon Nova family.

Discover Financial Services - Account Takeover (ATO) Fraud Strategy

Riverwoods, IL

Senior Data Science Analyst

Feb 2023 - Dec 2023

- Developed data-driven fraud detection and prevention strategies for account takeover (ATO) and Zelle fraud, using Python, SQL, and Snowflake to identify emerging attack patterns.
- Led fraud case reviews and deployed strategy updates that prevented hundreds of thousands of dollars in potential fraud losses.

Amazon - Alexa AI

Sunnyvale, CA

Applied Scientist Intern

May 2022 - Aug 2022

- Developed a coarse-to-fine reasoning framework for Visual Question Answering (VQA), combining question representations with Faster R-CNN image features for multimodal reasoning.
- Built an explainable VQA system with Transformer/LSTM-based explanation generation, producing natural-language rationales on VQA-E and GQA-REX without degrading VQA accuracy; resulting work was published at the NeurIPS 2022 TSRML Workshop.

Indian Institute of Science [SPIRE Lab]

Bengaluru, India

Junior Research Fellow (Guide: [Dr. Prasanta Kumar Ghosh](#))

Jul 2020 - Jul 2021

- Curated and validated multilingual speech datasets for the [MUCS 2021](#) Multilingual and Code-Switching ASR Challenge using Automatic Speech Recognition (ASR) metrics including Word Error Rate (WER) and acoustic likelihood.
- Trained a multilingual ASR baseline in Kaldi across six Indian languages, achieving ~30% Word Error Rate.
- Collaborated with Microsoft Research India, IBM, and Navana Tech on the MUCS 2021 challenge; resulting work was published at Interspeech 2021.

University of New Brunswick

Fredericton, Canada

MITACS Globalink Research Scholar (Guide: [Dr. Erik Scheme](#))

May 2019 - Aug 2019

- Developed a deep learning system for air-writing sentence recognition using electromyography (EMG) and inertial measurement unit (IMU) sensor data from a wrist-worn device, combining LSTM-based sequence modeling with CTC decoding.

- Developed machine learning models to detect inlet air-nozzle blockages from simulated aircraft sensor data, with Support Vector Machines (SVMs) achieving the best performance at 86% accuracy.

PUBLICATIONS & TECHNICAL REPORTS

(* denotes equal contribution)

Peer-Reviewed Publications

- Stress tests REVEAL fragile temporal and visual grounding in video-language models** [[Link](#)]
TV Sethuraman, Savya Khosla, Aditi Tiwari, Vidya Ganesh, Rakshana Jayaprakash, Aditya Jain, Vignesh Srinivasakumar, Onkar Kishor Susladkar, Srinidhi Sunkara, Aditya Shanmugham, Abbaas Alif Mohamed Nishar, [Rakesh Vaideeswaran Mahesh](#), Simon Jenni, Derek Hoiem
ICML 2026 Workshop on Mechanistic Interpretability | ICML 2026 Workshop on Combining Theory and Benchmarks
- Can the decoded text from automatic speech recognition effectively detect spoken grammar errors?** [[Link](#)]
Chowdam Venkata Thirumala Kumar, Meenakshi Sirigiraju, [Rakesh Vaideeswaran](#), Prasanta Kumar Ghosh, Chiranjeevi Yarra
SLaTE Workshop, Interspeech 2023
- Towards Reasoning-Aware Explainable VQA** [[Link](#)]
[Rakesh Vaideeswaran](#), Feng Gao, Abhinav Mathur, Govind Thattai
TSRML Workshop, NeurIPS 2022
- A study on native American English speech recognition by Indian listeners with varying word familiarity level** [[Link](#)]
Abhayjeet Singh, Achuth Rao, [Rakesh Vaideeswaran](#), Chiranjeevi Yarra, Prasanta Kumar Ghosh
O-COCOSDA 2021
- Multilingual and code-switching ASR challenges for low resource Indian languages** [[Link](#)]
Anuj Diwan*, [Rakesh Vaideeswaran](#)*, Sanket Shah*, Ankita Singh*, *et al.*
Interspeech 2021
- Analysis of machine learning algorithms for wi-fi-based indoor positioning system** [[Link](#)]
R Abishek*, KR Abishek*, N Hariharan*, [Rakesh Vaideeswaran](#)*, C Sundara Paripooranan*
IMICPW 2019

Technical Reports & Model Cards

- Amazon Nova 2: Multimodal reasoning and generation models** [[Link](#)]
Amazon Artificial General Intelligence (core contributor)
Amazon Science, 2025
- Amazon Nova Premier: Technical report and model card** [[Link](#)]
Amazon Artificial General Intelligence (core contributor)
Amazon Science, 2025
- The Amazon Nova Family of Models: Technical Report and Model Card** [[Link](#)]
Amazon Artificial General Intelligence (core contributor)
Amazon Science, 2024

PEER REVIEW & ACADEMIC SERVICE

- Reviewer, **AAAI Conference on Artificial Intelligence** - 2026 (5 papers), 2027 (4 papers)
- Reviewer, **ACM Conference on Recommender Systems (RecSys)** - 2026 (2 papers)
- Reviewer, **ICML Workshops** - 2026 (9 papers)
- Reviewer, **Amazon ML Conference (AMLC)** - 2025 (2 papers)
- Lead Organizer, **Multilingual and Code-Switching ASR Challenge** ([MUCS](#) workshop) - 2021 (20+ submissions)

TECHNICAL SKILLS

- Programming:** Python, C++, C, MATLAB, Bash, SQL
- ML / LLM Frameworks:** PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, OpenCV, Kaldi, vLLM

- **Infrastructure:** Git, Snowflake, Kubernetes, AWS (S3, EKS, EMR)
- **Research Areas:** Large Language Models (LLMs), Vision-Language Models (VLMs), foundation model pretraining and post-training, reinforcement learning, multimodal learning, agentic AI, model evaluation, factuality and grounding

EDUCATION

University of Illinois, Urbana-Champaign

Master of Engineering, Electrical and Computer Engineering

Focus Area: *Machine Learning and Signal Processing*

Dec 2022

GPA: 3.95/4

National Institute of Technology, Tiruchirappalli (NIT Trichy)

Bachelor of Technology, Electronics and Communication Engineering

Minor - *Computer Science and Engineering*

Jul 2020

GPA: 9.51/10

Relevant Coursework: Automatic Speech Recognition, Pattern Recognition, Random Processes, Computer Vision, Artificial Intelligence, Data Structures & Algorithms, Multimedia Signal Processing, Natural Language Processing

SELECTED PROJECTS

Dimension Estimation from Depth-Map of monocular Images [[Code](#), [Article](#)]

Dec 2019 - Mar 2020

- Trained a deep learning model based on Convolutional Autoencoder-Artificial Neural Network architecture to estimate the distance (in 3D space) between two arbitrary points on a 2D image, and achieved a Mean Deviation Error of 0.059 meters.

Indoor Location Utility Service (ILocUS) [[Code](#), [Working Video](#)]

Dec 2017 - Mar 2018

- Implemented an indoor positioning system for indoor navigation using Wi-Fi, with signal values from 3 NodeMCU modules using k-Nearest Neighbors Algorithm (75% accuracy).
- Secured 1st place (among 16 teams) in a Product Ideation event organized by Qualcomm.

EXTRACURRICULARS & LEADERSHIP

- **Rubik's Puzzle Coach at XOOG (now Famyio)** - One of the Top 4 coaches in May 2021 (among over 40 coaches); Mentored and coached over 20 kids aged 6-15 to solve the Rubik's Cube and motivated them to pursue advanced levels.
- Planned and Coordinated [Pragyan Cube Open 2019](#) (over 50 Participants) as the head of the organizing team, demonstrating leadership and teamwork; Organized [MUCS 2021 Workshop](#) (over 150 participants).